LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

'Rente mit 63'
Labour Force Participation among Eligible Older Workers in Germany

SPS Pareto Seminar - Research Projects in Empirical Economics

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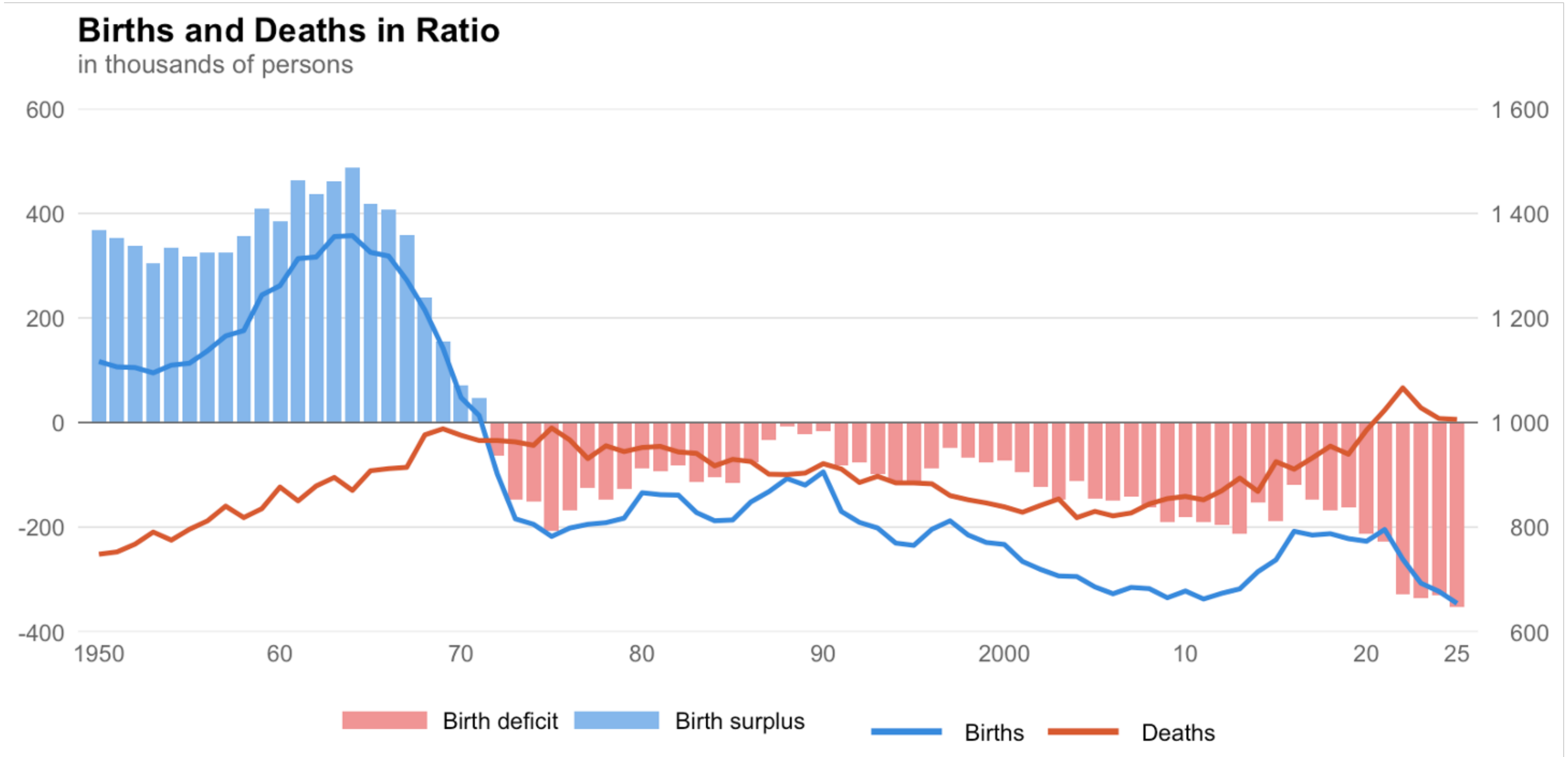
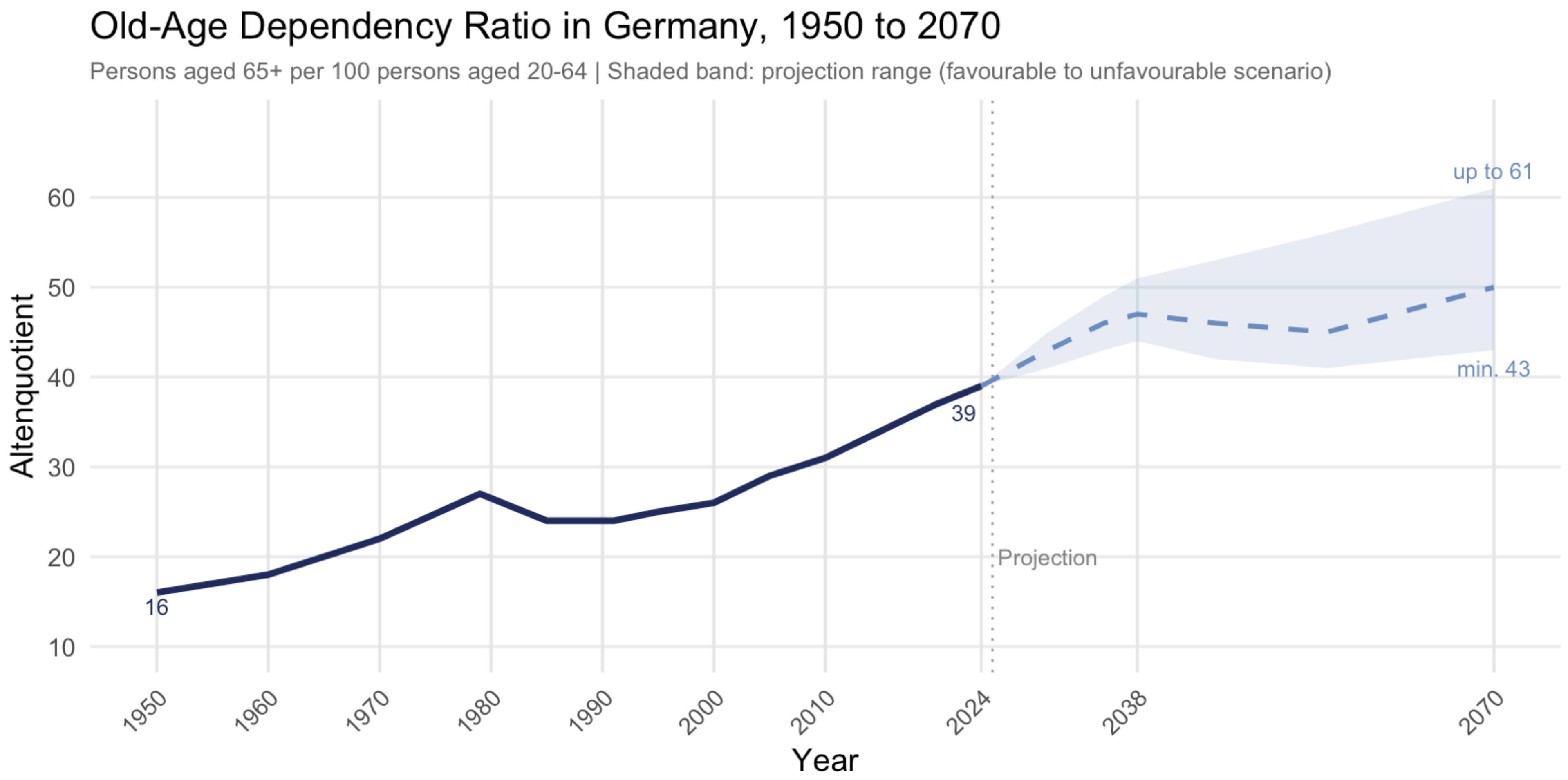
Germany Is Shrinking



Source: Statistisches Bundesamt (Destatis), 2026

The Demographic Burden Is Growing Fast



Source: Destatis, 16. koordinierte Bevölkerungsvorausberechnung (2025). Historical values: Destatis (2024).

How Germany's Pension System Works

- Pay-as-you-go
  - *Current workers finance current retirees*
- Early retirement penalty
  - *Permanent deduction of 0.3% per month for early retirement*
- Contribution years as eligibility
  - *45 qualifying years enable access to early retirement provisions*

Reform: Rente mit 63

- **Implemented**
  - *July 2014*
- **Eligibility**
  - *45+ contribution years*
- **Retirement age**
  - *reduced to 63, no deductions*

Birth Year	Earliest Retirement Age
before 1953	63 + 0 months
1953	63 + 2 months
1954	63 + 4 months
1955	63 + 6 months
1956	63 + 8 months
…	…
1964+	67

Source: Deutsche Rentenversicherung Bund

Research Question

MAIN RESEARCH QUESTION

Did the introduction of Rente mit 63 significantly reduce labour force participation among eligible workers?

Research Question Specifications

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RQ1 · REGIONAL HETEROGENEITY

East vs. West Germany

Does the effect vary between East and West Germany?

RQ2 · SOCIOECONOMIC HETEROGENEITY

RQ2a: Income

Does the effect differ across income groups?

RQ2b: Education

Does the effect differ across education groups?

Difference-in-Differences Framework

Main Regression

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \epsilon_{it}$$

RQ1 · Regional Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \beta_3 (Treat_i \times Post_t \times East_i) + \epsilon_{it}$$

Difference-in-Differences Framework

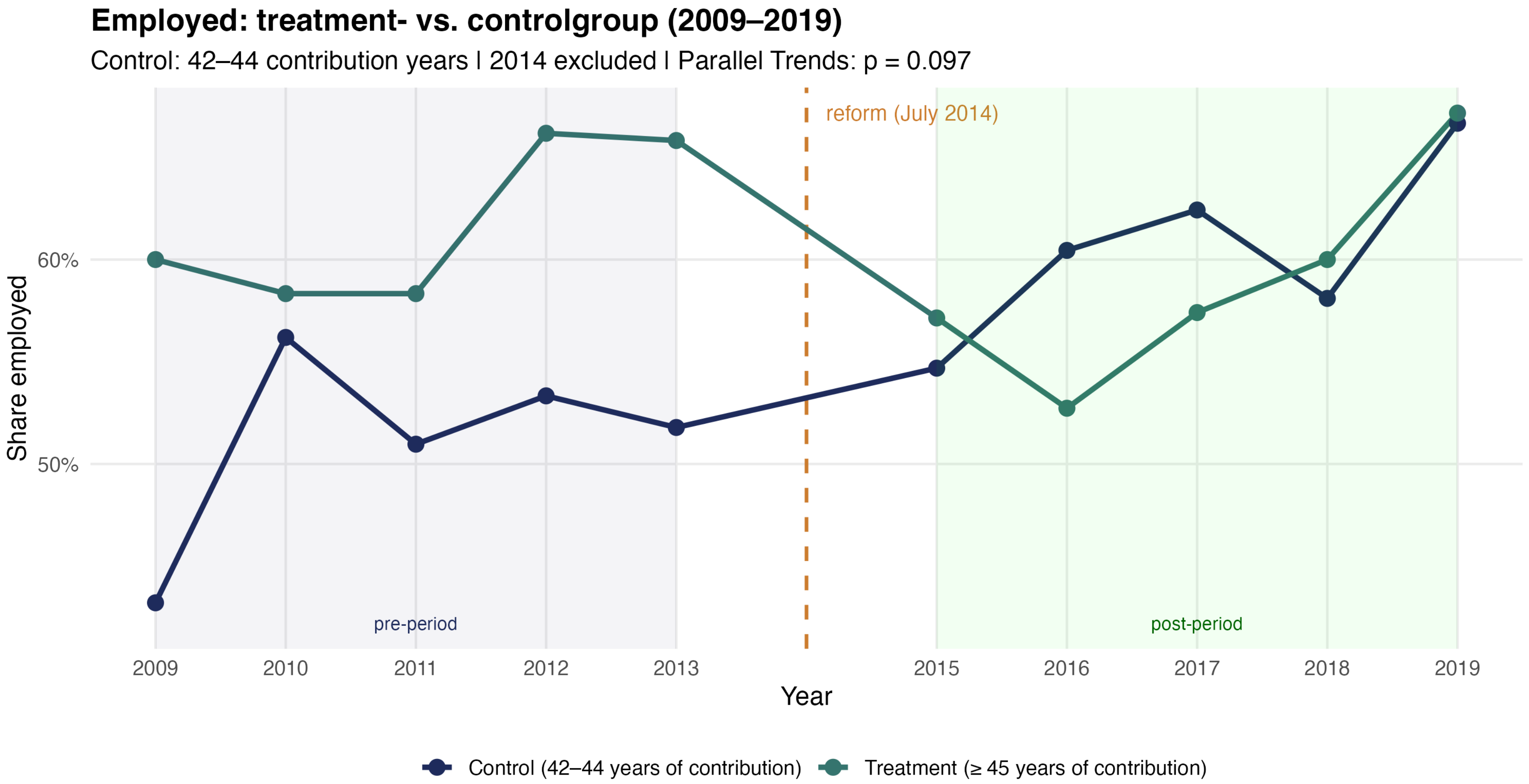
RQ2a · Income Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \sum_{i=1}^{4} \beta_i (Treat_i \times Post_t \times incq_i) + \epsilon_{it}$$

RQ2b · Educational Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \beta_3 \, (Treat_i \times Post_t \times HighEduc_i) + \epsilon_{it}$$

Control Group Selection



Source: SOEP-Data

Main Result

	Main Regression
Variables	
Post	-0.342^{***}
	(0.057)
Treat $\times$ Post	-0.219^{***}
	(0.050)
Observations	1,674
Individual FE	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001; SOEP-Data

Insignificant Heterogeneity

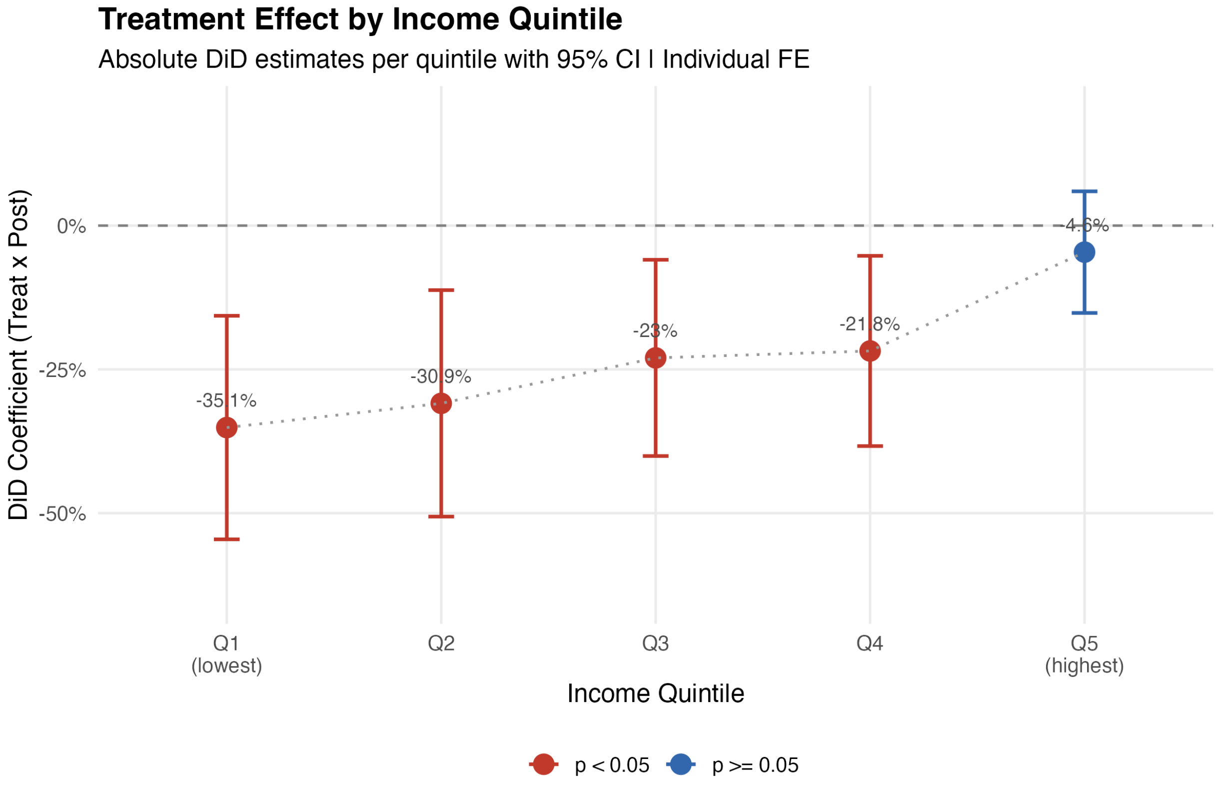
	RQ1: East/West	RQ2b: Education
Variables		
Post	-0.347^{***}	-0.342^{***}
	(0.056)	(0.057)
Treat × Post	-0.173^{***}	-0.212^{***}
	(0.052)	(0.058)
Treat × Post × East	−0.172	
	(0.111)	
Treat × Post × High Educ		−0.016
		(0.094)
Observations	1,674	1,674
Individual FE	Yes	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data

Significant Heterogeneity

	RQ2a: Income Gradient
Variables	
Post	-0.337^{***}
	(0.057)
Treat × Post (Q1 Ref.)	-0.351^{***}
	(0.099)
Treat × Post × Q2	+0.042
	(0.114)
Treat × Post × Q3	+0.121
	(0.123)
Treat × Post × Q4	+0.133
	(0.124)
Treat × Post × Q5	$+0.305^{**}$
	(0.110)
Observations	1,674
Individual FE	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data



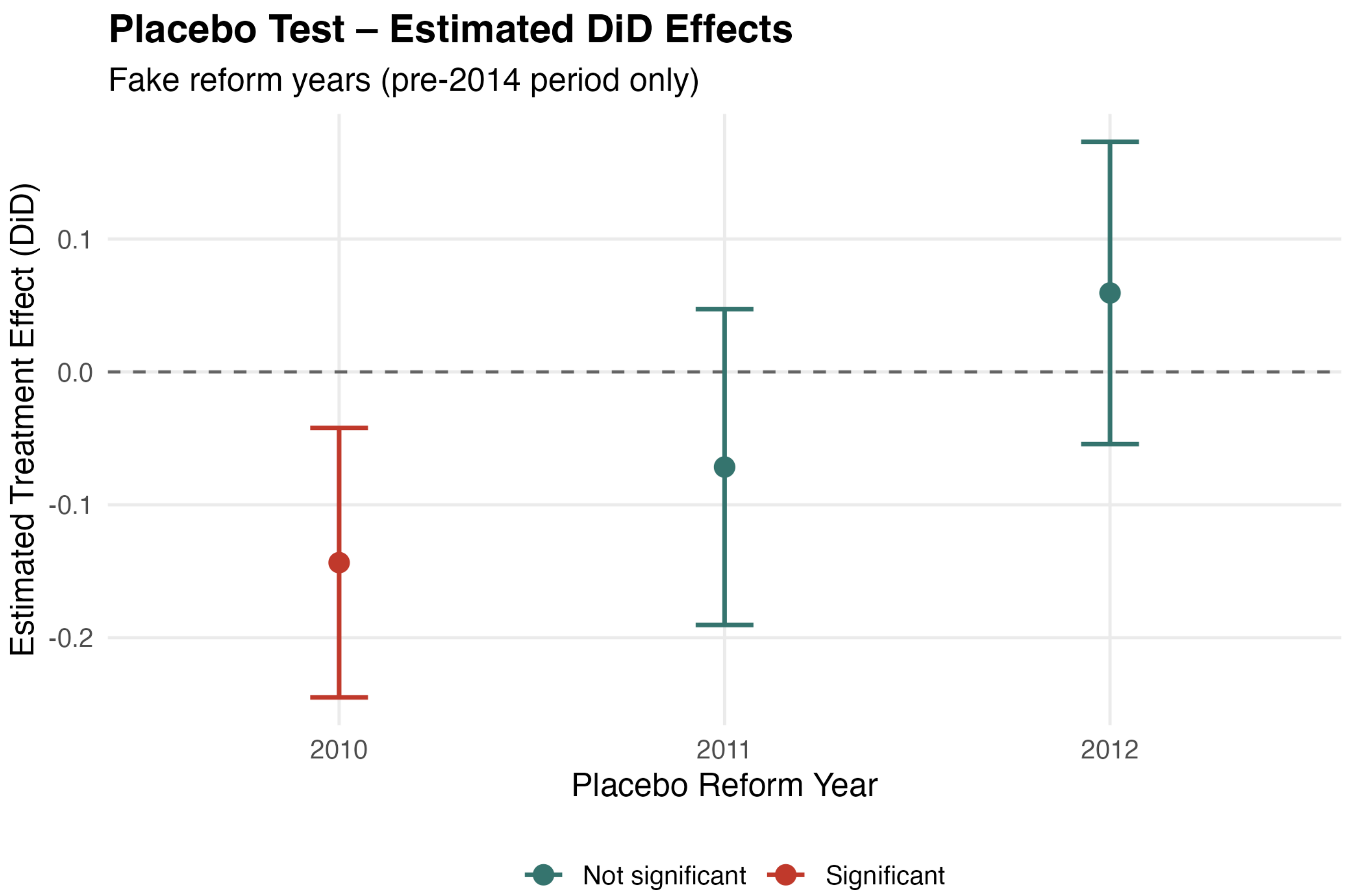
Source: SOEP-Data

Robustness: Alternative Control Groups

	35–44	40–44	41–44	42–44⋆
Variables				
Treat	0.239***	0.170***	0.139***	0.110*
	(0.038)	(0.040)	(0.041)	(0.043)
Post	0.136***	0.103***	0.108***	0.096**
	(0.018)	(0.025)	(0.028)	(0.032)
Treat × Post	−0.170***	−0.137*	−0.142*	−0.130*
	(0.053)	(0.055)	(0.056)	(0.058)
N (Total)	6,553	3,438	2,731	2,068
Ratio (C:T)	10.5:1	5.0:1	3.8:1	2.6:1

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data

Robustness: Placebo Tests



Source: SOEP-Data

Robustness: Donut Design

	Full Sample	Donut Sample
Variables		
Post	-0.342^{***}	-0.339^{***}
	(0.057)	(0.077)
Treat × Post	-0.219^{***}	-0.194^{**}
	(0.050)	(0.073)
Observations	1,674	1,232
Individual FE	Yes	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data

Robustness: Potential Confounder Flexi-Rente

	Baseline	Flexi-Dummy	2015–16 only
Variables			
Post	-0.342^{***}	-0.396^{***}	-0.410^{***}
	(0.057)	(0.057)	(0.064)
Treat × Post	-0.219^{***}	-0.114^{*}	-0.088
	(0.050)	(0.052)	(0.079)
Post-2017 (Flexirente)		-0.342^{***}	
		(0.045)	
Observations	1,674	1,674	1,072
Individual FE	Yes	Yes	Yes
Post-Period	2015–19	2015–19	2015–16

* p < 0.05, ** p < 0.01, *** p < 0.001 ; SOEP-Data

Limitations

- Men only
  - *Women rarely reach 45 years of contribution*
- Data
  - *Small treatment group*
  - *Approximated contribution years*
- Identification
  - *Flexirente effects post-2017*

Key Findings

- Rente mit 63 causally reduces employment
  - *by ~22 percentage points (DiD = -0.219***)*
- No uniform heterogeneity across subgroups
  - *region and education show no significant differential effects*
- Strong income gradient
  - *low earners (-35 pp) react most, top earners barely affected (-5 pp)*

Questions & Discussion

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Thank You!

for your attention. We look forward to your questions.

SLIDES & PAPER